Lecture 02:

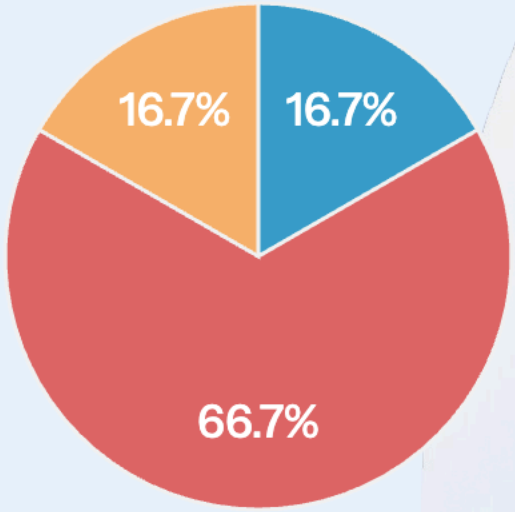
Ordinary Least Squares & Generalized Linear Models

Matthew Wicker

8 October 2026

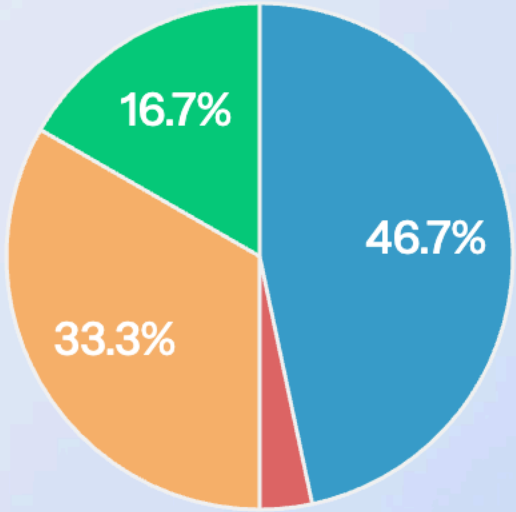
Thanks for the feedback!

Aspirations



- Academia/Research (pursue a PhD/Research Scientist)
- Industry (Research Engineer/Start up)
- Unsure/Non-specific goals

Worries

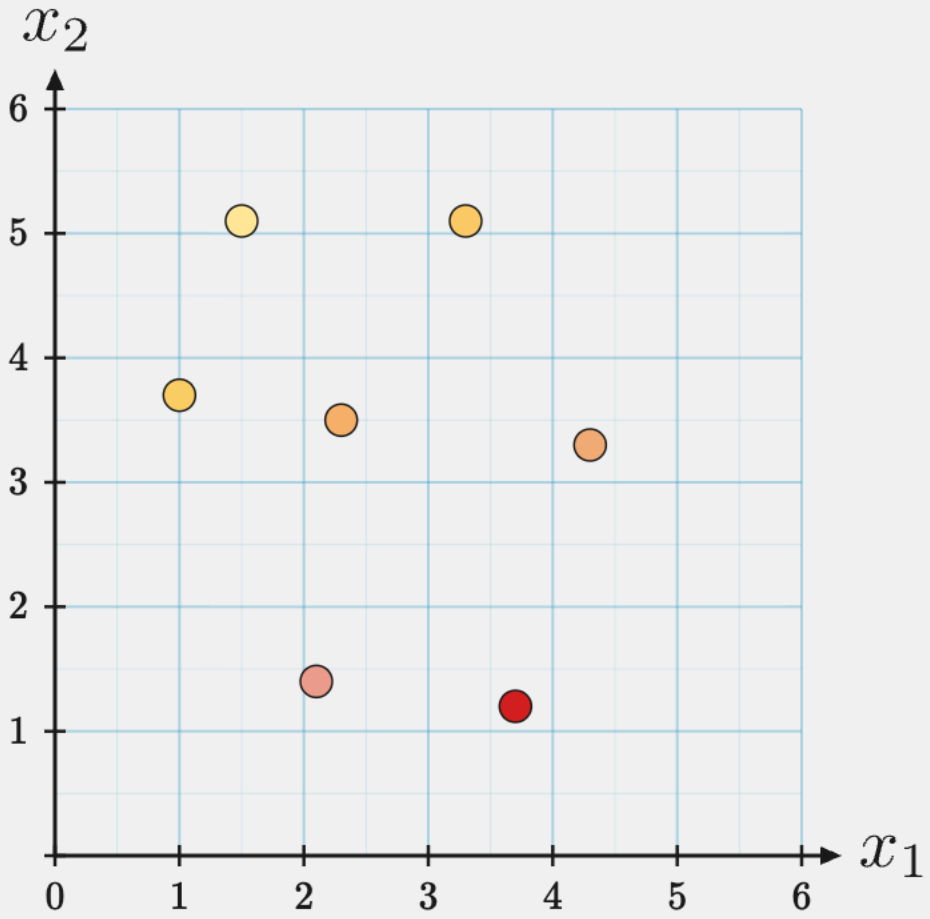


- Multivariate Calculus
- Linear Algebra
- Probability Theory
- None

At the one-hour mark we will have a 10-minute break, for all lectures that hit the hour mark.

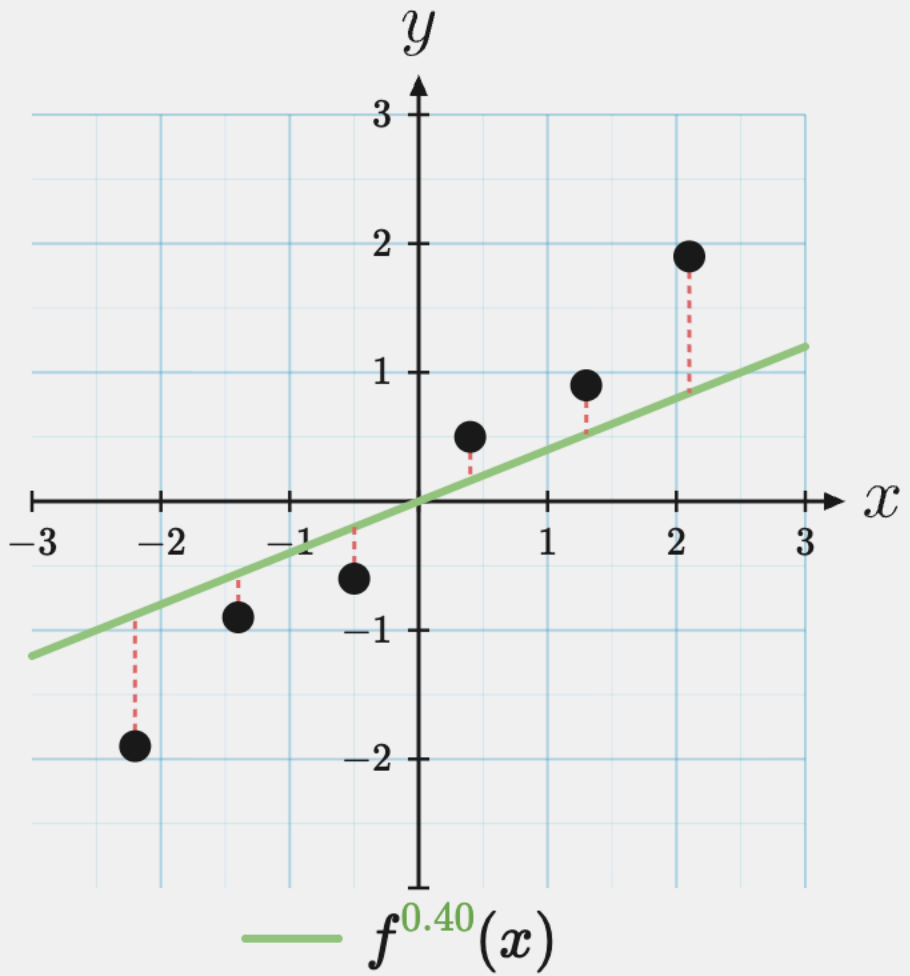
≈30 responses · released with ($\varepsilon = 1$, $\delta = 0.1$)-differential privacy (each chart)

The Linear Model



$$\boldsymbol{X} = \begin{bmatrix} 1.0 & 3.7 \\ 1.5 & 5.1 \\ 2.1 & 1.4 \\ 2.3 & 3.5 \\ 3.3 & 5.1 \\ 3.7 & 1.2 \\ 4.3 & 3.3 \end{bmatrix} \quad \boldsymbol{y} = \begin{bmatrix} 1.2 \\ 0.5 \\ 2.8 \\ 2.0 \\ 1.4 \\ 3.8 \\ 2.2 \end{bmatrix}$$

Measuring Error



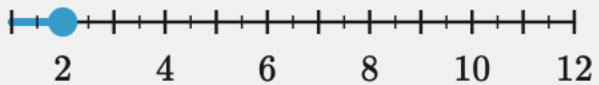
$$\hat{y} = \begin{bmatrix} -0.88 \\ -0.56 \\ -0.20 \\ 0.16 \\ 0.52 \\ 0.84 \end{bmatrix} \quad y = \begin{bmatrix} -1.9 \\ -0.9 \\ -0.6 \\ 0.5 \\ 0.9 \\ 1.9 \end{bmatrix} \quad \hat{y} - y = \begin{bmatrix} 1.02 \\ 0.34 \\ 0.40 \\ -0.34 \\ -0.38 \\ -1.06 \end{bmatrix}$$

$\|\hat{y}\|_p$	$\|y\|_p$	$\|\hat{y} - y\|_p$
1.459	3.074	1.643

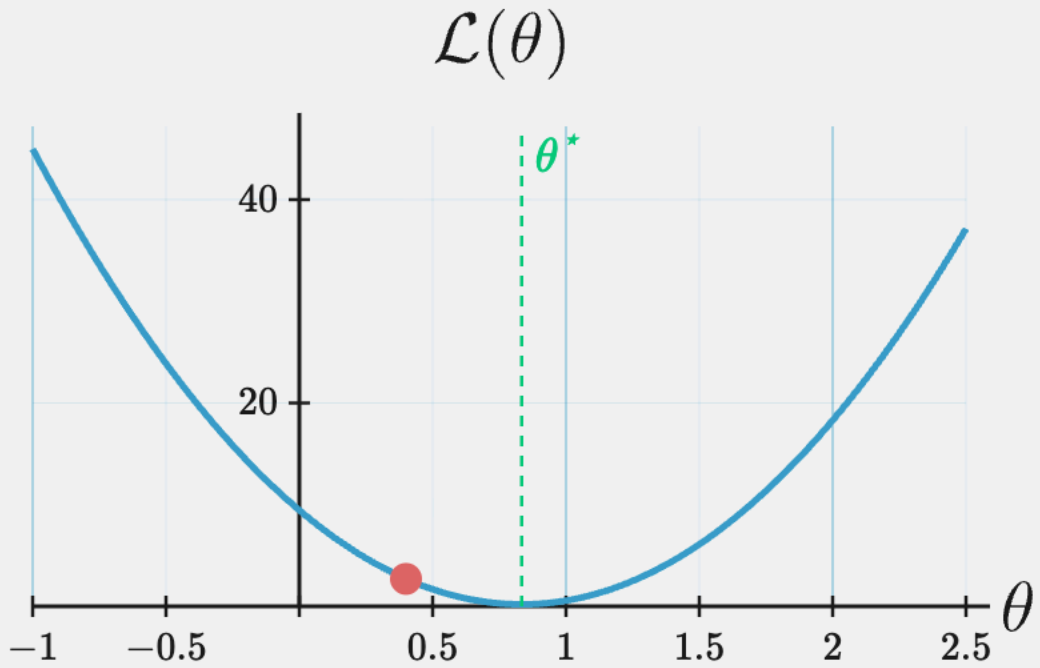
$w = 0.40$



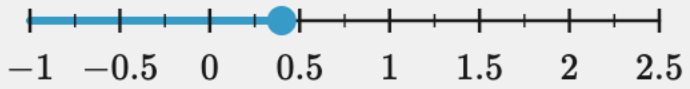
$p = 2.00$



The loss landscape



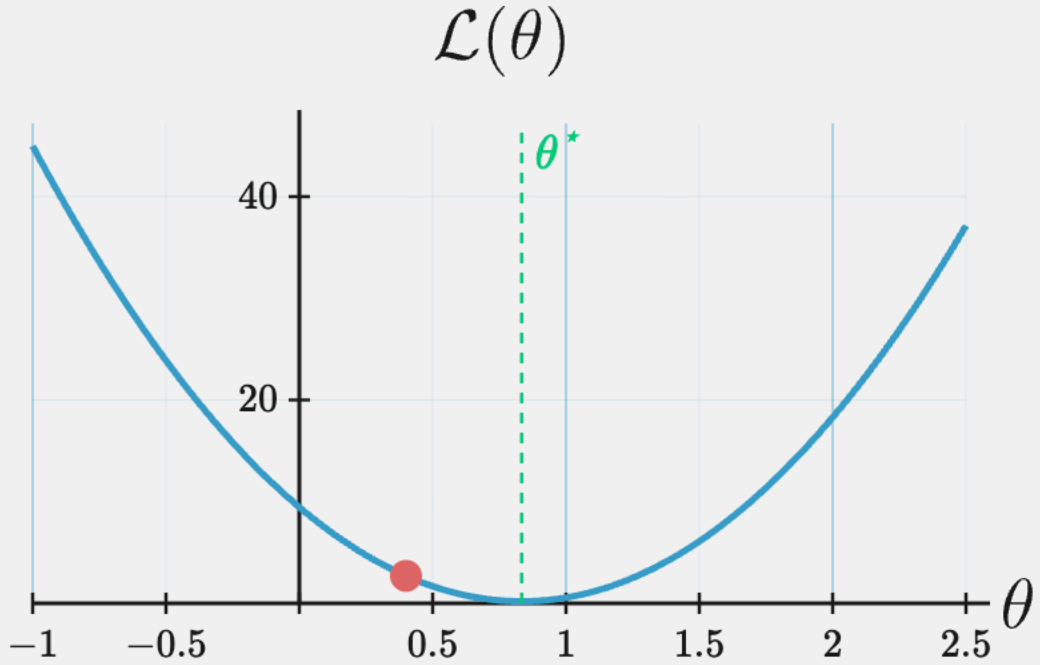
$\theta = 0.40$

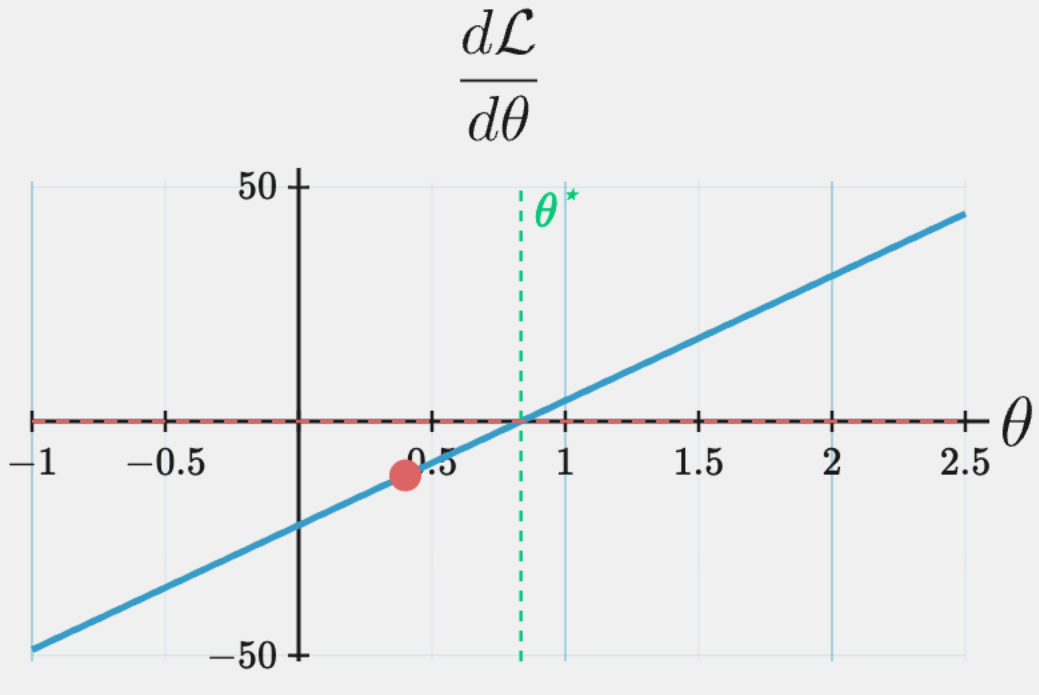


$\theta = 0.40, \quad \mathcal{L}(\theta) = 2.70$

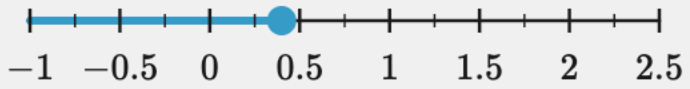
The loss landscape

The loss landscape will always achieve extrema where the gradient is equal to 0. We will explore this more systematically later, but is perhaps something you have seen in the past.





$\theta = 0.40$



$\theta = 0.40, \quad \mathscr{L}(\theta) = 2.70, \quad \mathrm{d}\mathscr{L}/\mathrm{d}\theta = -11.55$

Our first learning algorithm: direct solution

We have our loss for linear regression with mean squared error is:

$$\mathcal{L}(\theta) = \|\boldsymbol{X}\theta - \boldsymbol{y}\|_2^2$$

1. Expand our loss function
2. Take gradient of loss function
3. Set gradient equal to zero, solve for theta

Expanding our loss function

In order to take the derivative of this loss function, we will need to deal with the fact that it is a function involving vectors and matrices

$$\mathcal{L}(\boldsymbol{y}, \hat{\boldsymbol{y}}) = \sum_i \left(y_i - \sum_j X_{i,j} \theta_j \right)^2$$

$$= (\boldsymbol{X}\theta - \boldsymbol{y})^\top (\boldsymbol{X}\theta - \boldsymbol{y})$$

Expanding our loss function

Let's further expand the loss to make it easier to differentiate:

$$(\boldsymbol{X}\theta - \boldsymbol{y})^\top (\boldsymbol{X}\theta - \boldsymbol{y})$$

$$= (\theta^\top \boldsymbol{X}^\top - \boldsymbol{y}^\top)(\boldsymbol{X}\theta - \boldsymbol{y})$$

$(\boldsymbol{X}\theta - \boldsymbol{y})^\top = \theta^\top \boldsymbol{X}^\top - \boldsymbol{y}^\top$

$$= \theta^\top \boldsymbol{X}^\top \boldsymbol{X}\theta - \theta^\top \boldsymbol{X}^\top \boldsymbol{y} - \boldsymbol{y}^\top \boldsymbol{X}\theta + \boldsymbol{y}^\top \boldsymbol{y}$$

multiply out: four terms

$$= \theta^\top (\boldsymbol{X}^\top \boldsymbol{X})\theta - 2\,\boldsymbol{y}^\top \boldsymbol{X}\theta + \boldsymbol{y}^\top \boldsymbol{y}$$

$\theta^\top \boldsymbol{X}^\top \boldsymbol{y}$ is a scalar, so it equals its transpose $\boldsymbol{y}^\top \boldsymbol{X}\theta$: the middle terms combine

Differentiate each component

Now we need to take the derivative, so let's differentiate each component. For some, this will involve recalling identities, for others, you may need to prove the identity for yourself a few times first before you feel comfortable.

$$\left(\theta^\top(\boldsymbol{X}^\top\boldsymbol{X})\theta - 2(\boldsymbol{y}^\top\boldsymbol{X}\theta) + \boldsymbol{y}^\top\boldsymbol{y}\right)$$

Introduction to Vector Calculus

As a course convention, we will state all gradients in numerator format. This means that a function $f:\mathbb{R}^n \to \mathbb{R}^m$ has gradient in $\mathbb{R}^{m\times n}$. We will begin with what we will term *basic* vector calculus which follows as direct analogues of single variable calculus.

Rule	**Single-variable**	**Vector (numerator layout)**
Constant	$\frac{d}{dx}a = 0, \quad \frac{d}{dx}(af) = af'$	$\nabla_\theta a = \mathbf{0}^\top, \quad \nabla_\theta(af) = a\nabla_\theta f, \quad \nabla_\theta \boldsymbol{c}^\top\theta = \boldsymbol{c}^\top$
Sum	$(f+g)' = f' + g'$	$\nabla_\theta(f+g) = \nabla_\theta f + \nabla_\theta g$
Product	$(fg)' = f'g + fg'$	$\nabla_\theta(fg) = g\nabla_\theta f + f\nabla_\theta g$
Power	$(x^p)' = px^{p-1}$	$\nabla_\theta f^p = pf^{p-1}\nabla_\theta f, \quad \nabla_\theta(\theta^\top\theta) = 2\theta^\top$

Note: only for inner-product, not outer-product, which will be covered later.

Proving vector calc. identities

$$\nabla_\theta \boldsymbol{c}^\top \theta = \boldsymbol{c}^\top$$

In this course we will use numerator layout differentiation, this means that the gradient of a vector valued function is a row vector.

For more information on this convention please see:

Page 146 and 147 of the MML book

Proving vector calc. identities

$$\nabla_\theta \boldsymbol{c}^\top \theta = \boldsymbol{c}^\top$$

Some general steps for you to consider:

1) Break things into summation notation

2) Differentiate w.r.t. an arbitrary element

3) Assemble into the required shape

Proving vector calc. identities

$$\nabla_\theta \boldsymbol{c}^\top \theta = \boldsymbol{c}^\top$$

Proof:

$$\boldsymbol{c}^\top \theta = \sum_j c_j \theta_j$$

$$\frac{\partial \boldsymbol{c}^\top \theta}{\partial \theta_j} = c_j$$

$$\nabla_\theta \boldsymbol{c}^\top \theta = \boldsymbol{c}^\top$$

Proving vector calc. identities

$$\nabla_\theta(\theta^\top \boldsymbol{A}\theta) = \theta^\top(\boldsymbol{A} + \boldsymbol{A}^\top)$$

$$\theta^\top \boldsymbol{A}\theta = \sum_i \sum_j \theta_i \theta_j A_{i,j},$$

$$\frac{\partial(\theta^\top \boldsymbol{A}\theta)}{\partial \theta_k} = \sum_j A_{k,j}\theta_j + \sum_i \theta_i A_{i,k} = (\boldsymbol{A}\theta)_k + (\boldsymbol{A}^\top\theta)_k,$$

$$\nabla_\theta(\theta^\top \boldsymbol{A}\theta) = \theta^\top(\boldsymbol{A} + \boldsymbol{A}^\top).$$

Vector calc. identities to practice proving

$$\frac{\partial}{\partial \boldsymbol{v}} \boldsymbol{u}^\top \boldsymbol{v} = \boldsymbol{u}^\top$$

$$\frac{\partial}{\partial \boldsymbol{u}} \boldsymbol{u}^\top \boldsymbol{v} = \boldsymbol{v}^\top$$

$$\frac{\partial}{\partial \boldsymbol{v}} \boldsymbol{A}\boldsymbol{v} = \boldsymbol{A}$$

$$\frac{\partial}{\partial \boldsymbol{v}} \boldsymbol{v}^\top \boldsymbol{A} = \boldsymbol{A}^\top$$

$$\nabla_\theta (\theta^\top \boldsymbol{A} \theta) = \theta^\top (\boldsymbol{A} + \boldsymbol{A}^\top)$$

Returning to the loss function

Using the rules we just proved:

$$\nabla_\theta(\theta^\top \boldsymbol{A}\theta) = \theta^\top(\boldsymbol{A} + \boldsymbol{A}^\top) \qquad \nabla_\theta \boldsymbol{c}^\top \theta = \boldsymbol{c}^\top$$

Taking the derivative of our function seems more manageable.

$$\mathcal{L} = \theta^\top(\boldsymbol{X}^\top \boldsymbol{X})\theta - 2(\boldsymbol{y}^\top \boldsymbol{X}\theta) + \boldsymbol{y}^\top \boldsymbol{y}$$

$$\nabla_\theta \mathcal{L} =$$

Returning to the loss function

Using the rules we just proved:

$$\nabla_\theta(\theta^\top \boldsymbol{A}\theta) = \theta^\top(\boldsymbol{A} + \boldsymbol{A}^\top) \qquad \nabla_\theta \boldsymbol{c}^\top\theta = \boldsymbol{c}^\top$$

Taking the derivative of our function seems more manageable.

$$\mathcal{L} = \theta^\top(\boldsymbol{X}^\top\boldsymbol{X})\theta - 2(\boldsymbol{y}^\top\boldsymbol{X}\theta) + \boldsymbol{y}^\top\boldsymbol{y}$$

$$\nabla_\theta\mathcal{L} = 2\Big(\theta^\top(\boldsymbol{X}^\top\boldsymbol{X}) - \boldsymbol{y}^\top\boldsymbol{X}\Big)$$

To find minimum, let's set equal to 0

$$\nabla_\theta \mathcal{L} = 2\Big(\theta^\top (\boldsymbol{X}^\top \boldsymbol{X}) - \boldsymbol{y}^\top \boldsymbol{X}\Big)$$

$$0 = 2\Big(\theta^\top (\boldsymbol{X}^\top \boldsymbol{X}) - \boldsymbol{y}^\top \boldsymbol{X}\Big)$$

set the gradient to zero

$$\theta^\top (\boldsymbol{X}^\top \boldsymbol{X}) = \boldsymbol{y}^\top \boldsymbol{X}$$

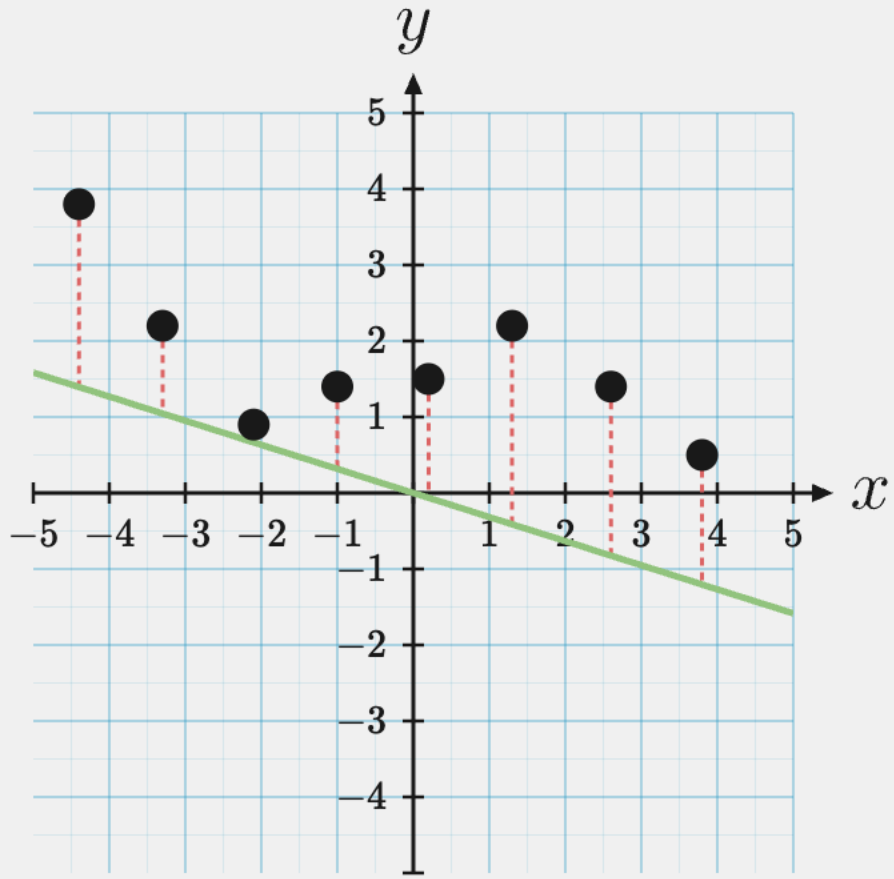
divide by 2; move $\boldsymbol{y}^\top \boldsymbol{X}$ across

$$\theta = (\boldsymbol{X}^\top \boldsymbol{X})^{-1} \boldsymbol{X}^\top \boldsymbol{y}$$

transpose ($\boldsymbol{X}^\top \boldsymbol{X}$ is symmetric), multiply by $(\boldsymbol{X}^\top \boldsymbol{X})^{-1}$

Ordinary Least Squares (OLS) Estimate

Now we have a solution to the best line to fit this data!



$$X = \begin{bmatrix} -4.4 \\ -3.3 \\ -2.1 \\ -1.0 \\ 0.2 \\ 1.3 \\ 2.6 \\ 3.8 \end{bmatrix} \quad y = \begin{bmatrix} 3.8 \\ 2.2 \\ 0.9 \\ 1.4 \\ 1.5 \\ 2.2 \\ 1.4 \\ 0.5 \end{bmatrix}$$

$$\theta = (X^\top X)^{-1} X^\top y$$

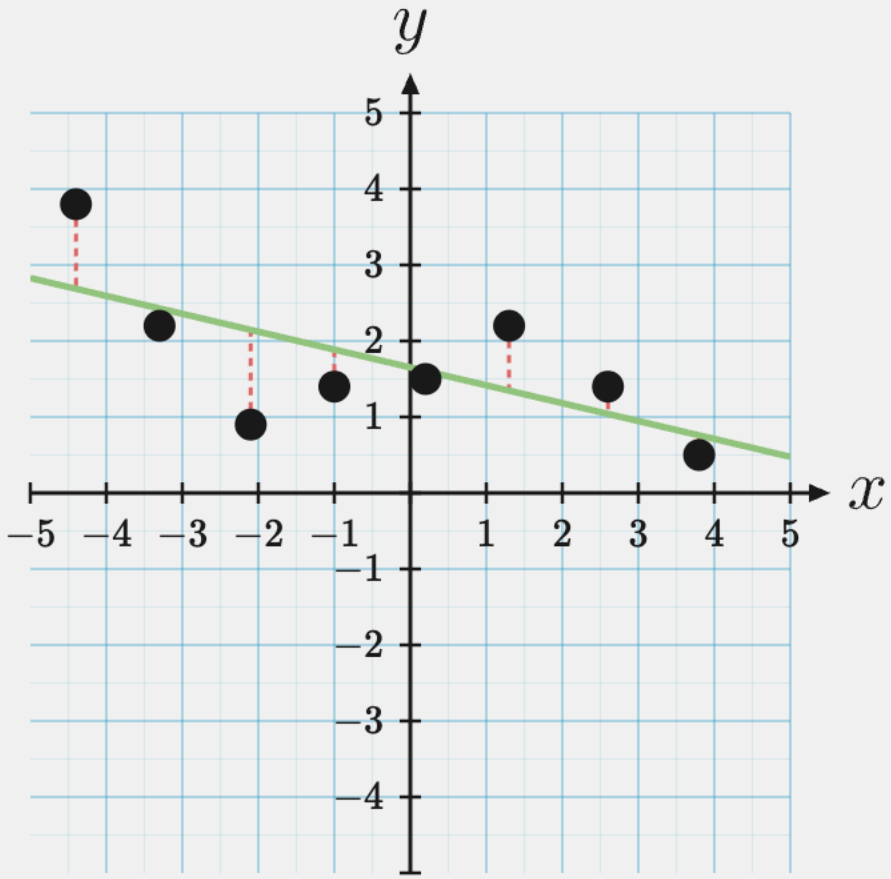
$$\theta = -0.32$$

$c = 0.0$



As I have drawn it here, this seems like a powerful result, but for a weak model

OLS with an intercept



$$\boldsymbol{X} = \begin{bmatrix} -4.4 \\ -3.3 \\ -2.1 \\ -1.0 \\ 0.2 \\ 1.3 \\ 2.6 \\ 3.8 \end{bmatrix} \quad \boldsymbol{y} = \begin{bmatrix} 3.8 \\ 2.2 \\ 0.9 \\ 1.4 \\ 1.5 \\ 2.2 \\ 1.4 \\ 0.5 \end{bmatrix}$$

$$\hat{y} = wx + b$$

$$w = ???, \qquad b = ???$$

$c = 0.0$



Basis Expansion

All of the steps we have done so far work as long as the feature-space and output space are related by a linear transformation.

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> **Definition 1.3.1. Linear transformation** A linear transformation between two vector spaces V and W is a map $T : V \to W$ such that the following hold:
>
> - $T(\boldsymbol{v}_1 + \boldsymbol{v}_2) = T(\boldsymbol{v}_1) + T(\boldsymbol{v}_2)$ for any vectors $\boldsymbol{v}_1$ and $\boldsymbol{v}_2$ in V, and
> - $T(\alpha \boldsymbol{v}) = \alpha T(\boldsymbol{v})$ for any scalar α.

Try it: Prove that $y = mx + b$ is ***not*** linear in x.

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- $T(\alpha \boldsymbol{v}) = \alpha T(\boldsymbol{v})$ for any scalar α.

Try it: Prove that $y = mx + b$ is ***not*** linear in x.

- Let $f(x) = mx + b$ for any $m \in \mathbb{R}$, $b = 1$
- let $a = 0 \Rightarrow f(0) = 1$
- This violates the second property above

Basis Expansion

Though $y = mx + b$ does not relate x and y via a linear transformation, it can be expressed as a linear transformation from an *expanded basis*

$$\text{Let } x \in \mathbb{R} \qquad \phi(x) := [x, 1]^\top$$

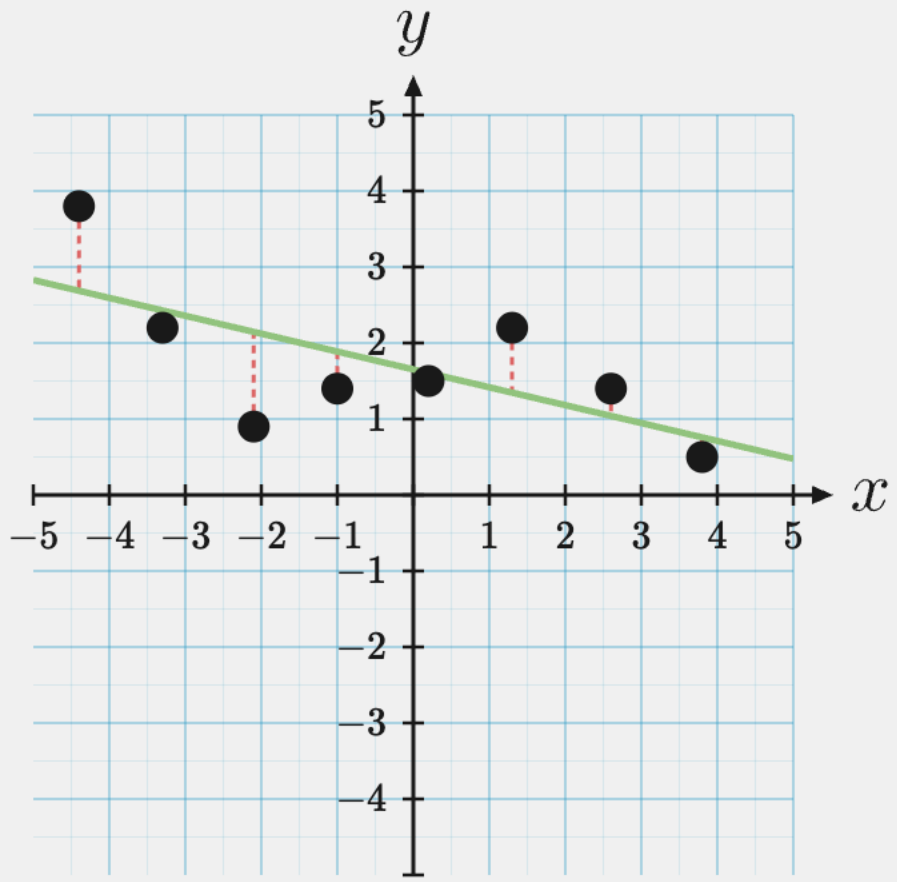
$$\hat{y} = \phi(x)^\top \theta$$

Affine Basis Expansion

$$\text{Let } x \in \mathbb{R} \qquad \phi(x) := [x, 1]^\top$$

$$\begin{aligned}
\hat{y} &= \phi(x)^\top \theta \\
&= [x, 1]\,[\theta_1, \theta_2]^\top \\
&= x\theta_1 + 1\theta_2 \\
&= x\theta_1 + \theta_2 \\
&= mx + b
\end{aligned}$$

Affine Basis Expansion



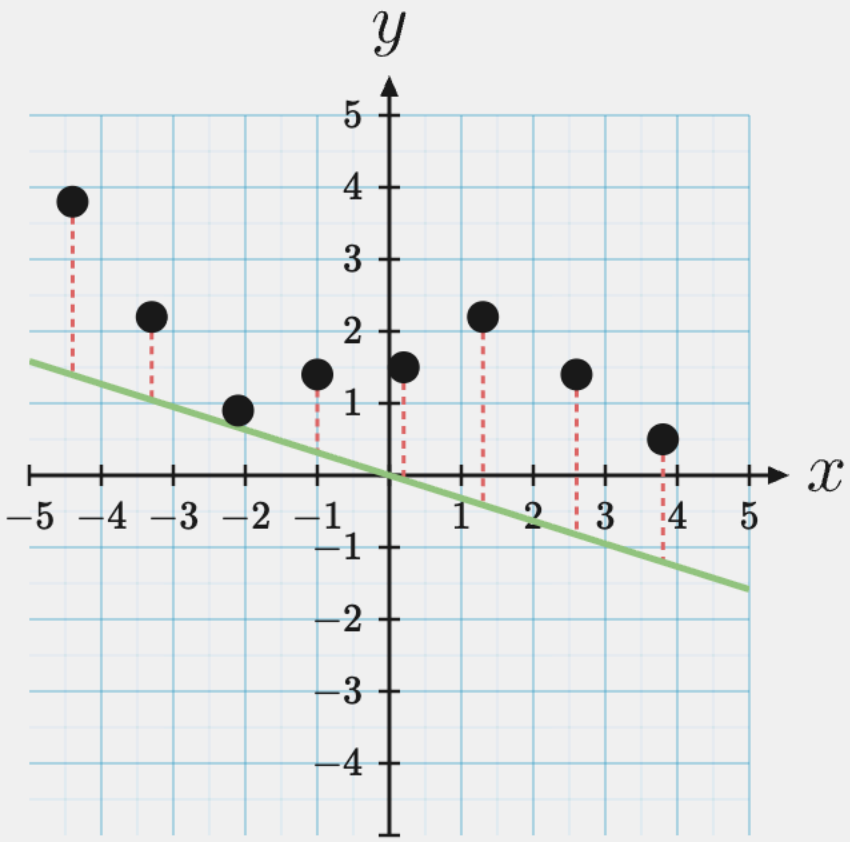
$$\phi(\boldsymbol{X}) = \begin{bmatrix} -4.4 & 1 \\ -3.3 & 1 \\ -2.1 & 1 \\ -1.0 & 1 \\ 0.2 & 1 \\ 1.3 & 1 \\ 2.6 & 1 \\ 3.8 & 1 \end{bmatrix} \quad \boldsymbol{y} = \begin{bmatrix} 3.8 \\ 2.2 \\ 0.9 \\ 1.4 \\ 1.5 \\ 2.2 \\ 1.4 \\ 0.5 \end{bmatrix}$$

$$\theta = \left(\phi(\boldsymbol{X})^\top \phi(\boldsymbol{X})\right)^{-1} \phi(\boldsymbol{X})^\top \boldsymbol{y}$$

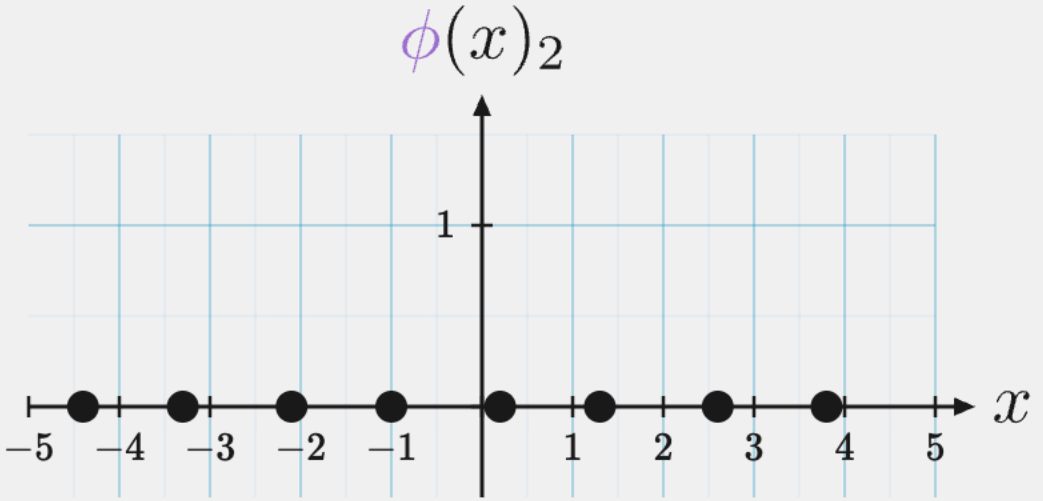
$$\theta = [-0.24,\ 1.65]^\top$$

Basis expansion visualization

OLS without basis expansion

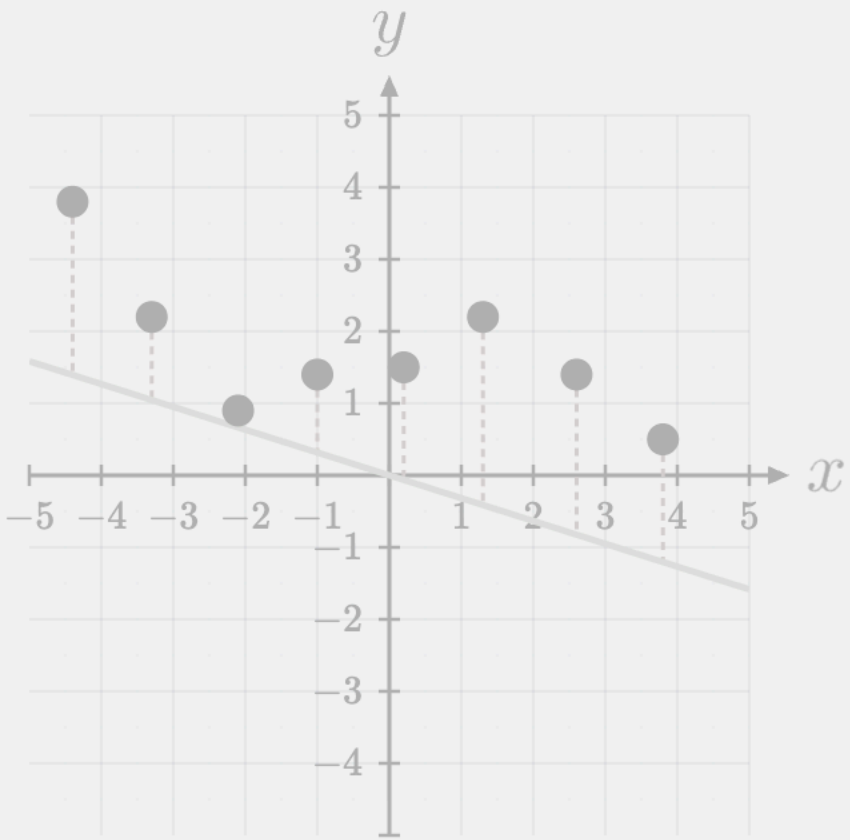


The inputs only (no labels)

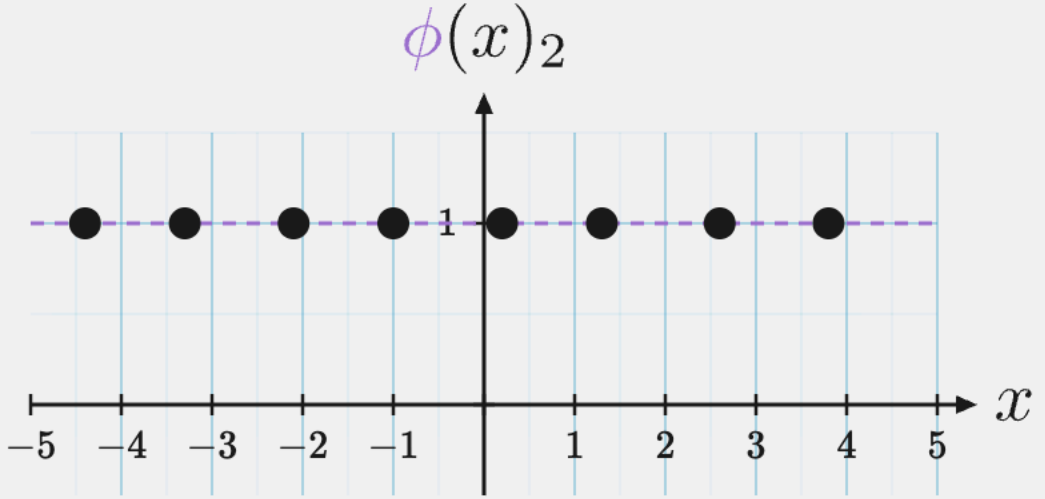


Basis expansion visualization

OLS without basis expansion

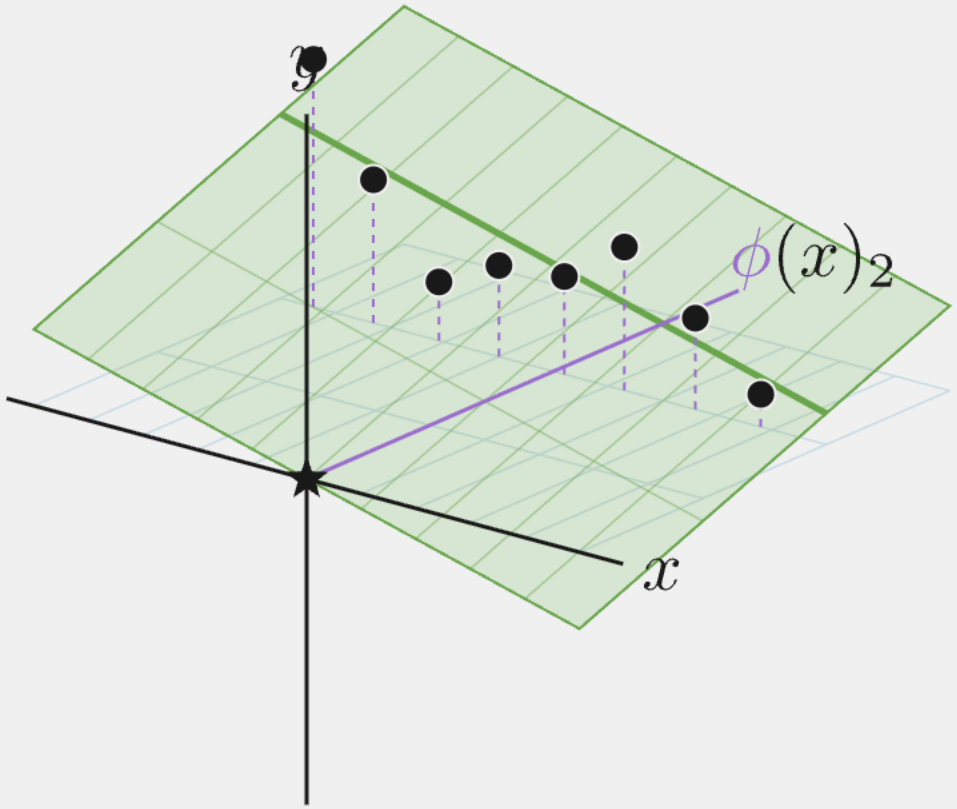


The inputs in the expanded basis $\varphi(x) = [x, 1]^\top$

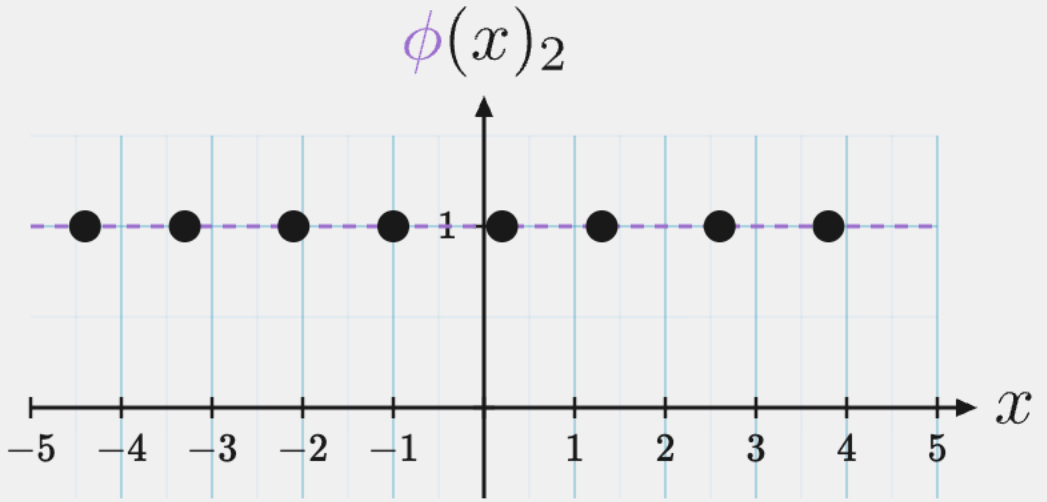


Basis expansion visualization

OLS in the basis-expanded space

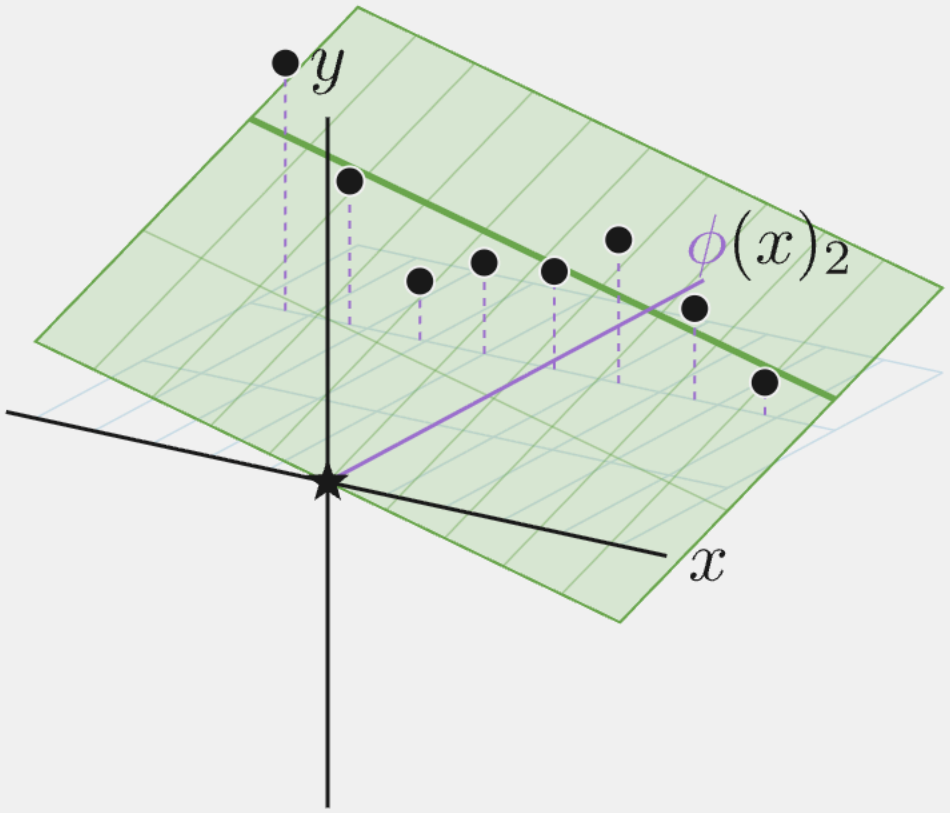


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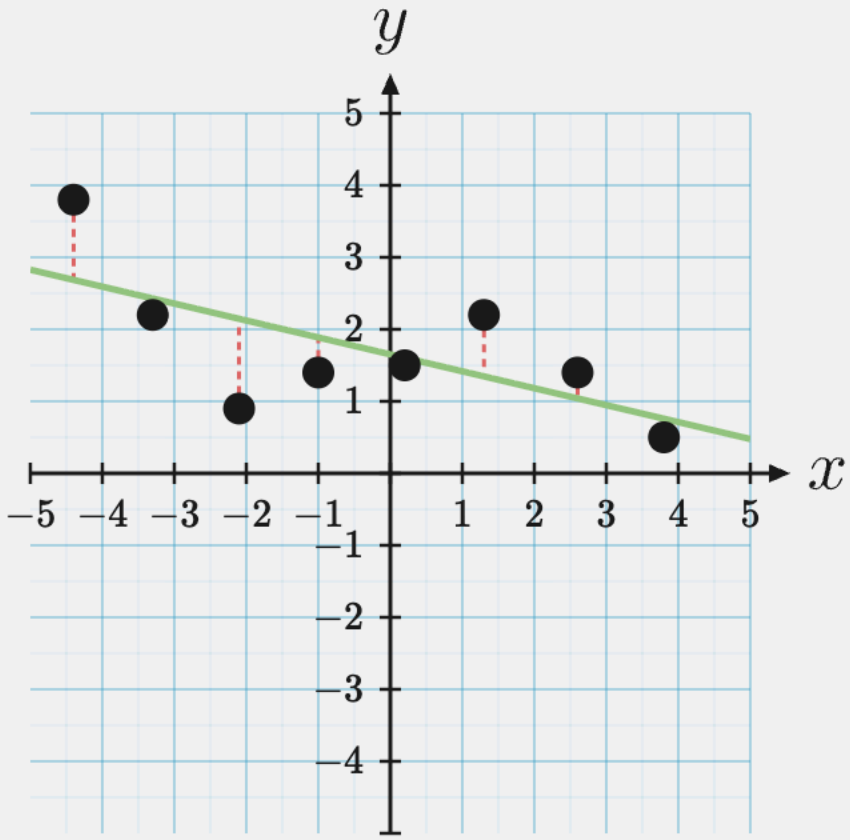


Basis expansion visualization

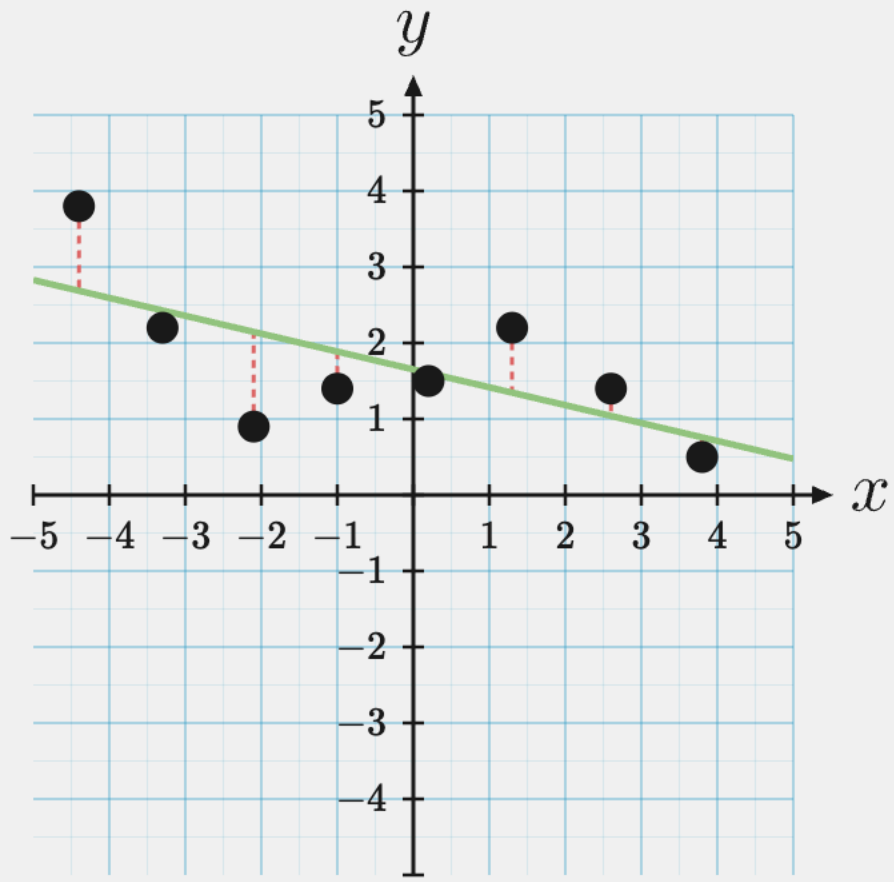
OLS in the basis-expanded space



OLS projected back to the data space



OLS with an intercept



$$\phi(\boldsymbol{X}) = \begin{bmatrix} -4.4 & 1 \\ -3.3 & 1 \\ -2.1 & 1 \\ -1.0 & 1 \\ 0.2 & 1 \\ 1.3 & 1 \\ 2.6 & 1 \\ 3.8 & 1 \end{bmatrix} \quad \boldsymbol{y} = \begin{bmatrix} 3.8 \\ 2.2 \\ 0.9 \\ 1.4 \\ 1.5 \\ 2.2 \\ 1.4 \\ 0.5 \end{bmatrix}$$

$$\hat{y} = wx + b$$

$$[w, b]^\top = \left(\phi(\boldsymbol{X})^\top \phi(\boldsymbol{X})\right)^{-1} \phi(\boldsymbol{X})^\top \boldsymbol{y}$$

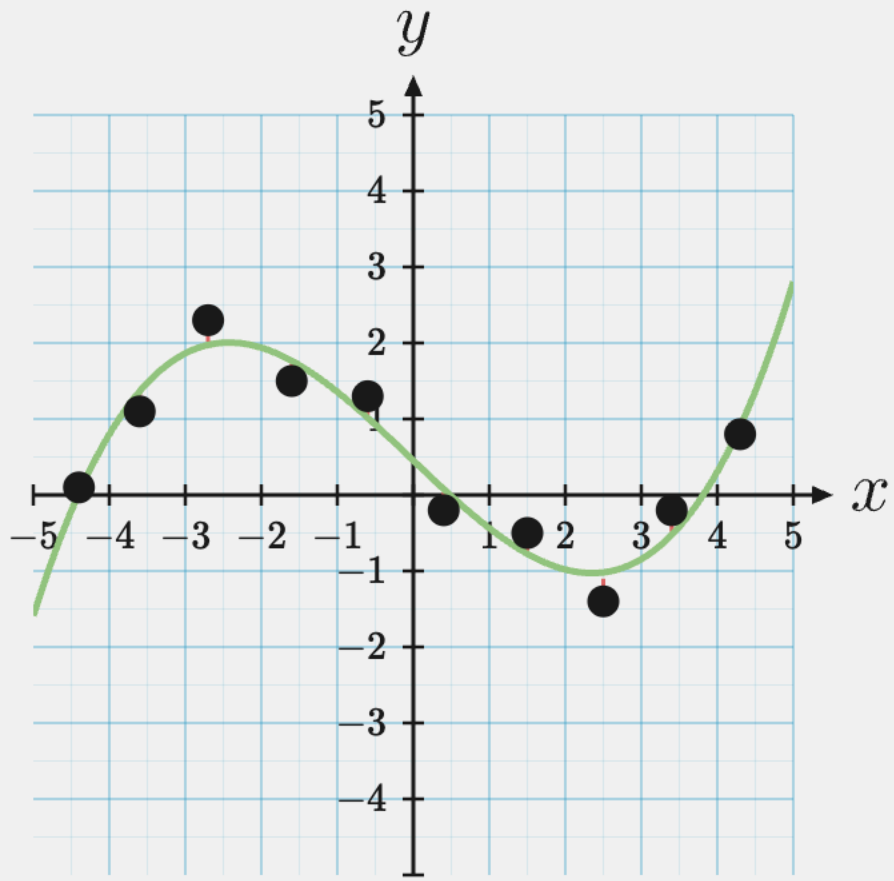
$$w = -0.24, \quad b = 1.65$$

Polynomial Basis Expansion

$$\text{Let } x \in \mathbb{R} \qquad \phi(x) := [x^3, x^2, x, 1]^\top$$

$$\begin{aligned}
\hat{y} &= \phi(x)^\top \theta \\
&= \left[x^3, x^2, x, 1\right] \left[\theta_1, \theta_2, \theta_3, \theta_4\right]^\top \\
&= x^3\theta_1 + x^2\theta_2 + x\theta_3 + 1\theta_4 \\
&= x^3\theta_1 + x^2\theta_2 + x\theta_3 + \theta_4
\end{aligned}$$

Polynomial Basis Expansion



$$\phi(\boldsymbol{X}) = \begin{bmatrix} -85.2 & 19.4 & -4.4 & 1 \\ -46.7 & 13.0 & -3.6 & 1 \\ -19.7 & 7.3 & -2.7 & 1 \\ -4.1 & 2.6 & -1.6 & 1 \\ -0.2 & 0.4 & -0.6 & 1 \\ 0.1 & 0.2 & 0.4 & 1 \\ 3.4 & 2.3 & 1.5 & 1 \\ 15.6 & 6.3 & 2.5 & 1 \\ 39.3 & 11.6 & 3.4 & 1 \\ 79.5 & 18.5 & 4.3 & 1 \end{bmatrix} \quad \boldsymbol{y} = \begin{bmatrix} 0.1 \\ 1.1 \\ 2.3 \\ 1.5 \\ 1.3 \\ -0.2 \\ -0.5 \\ -1.4 \\ -0.2 \\ 0.8 \end{bmatrix}$$

$$\theta = \left(\phi(\boldsymbol{X})^\top \phi(\boldsymbol{X})\right)^{-1} \phi(\boldsymbol{X})^\top \boldsymbol{y}$$

$$\theta = [0.06,\ 0.01,\ -0.95,\ 0.45]^\top$$

Polynomial Basis Expansion

$$\text{Let } x \in \mathbb{R}$$

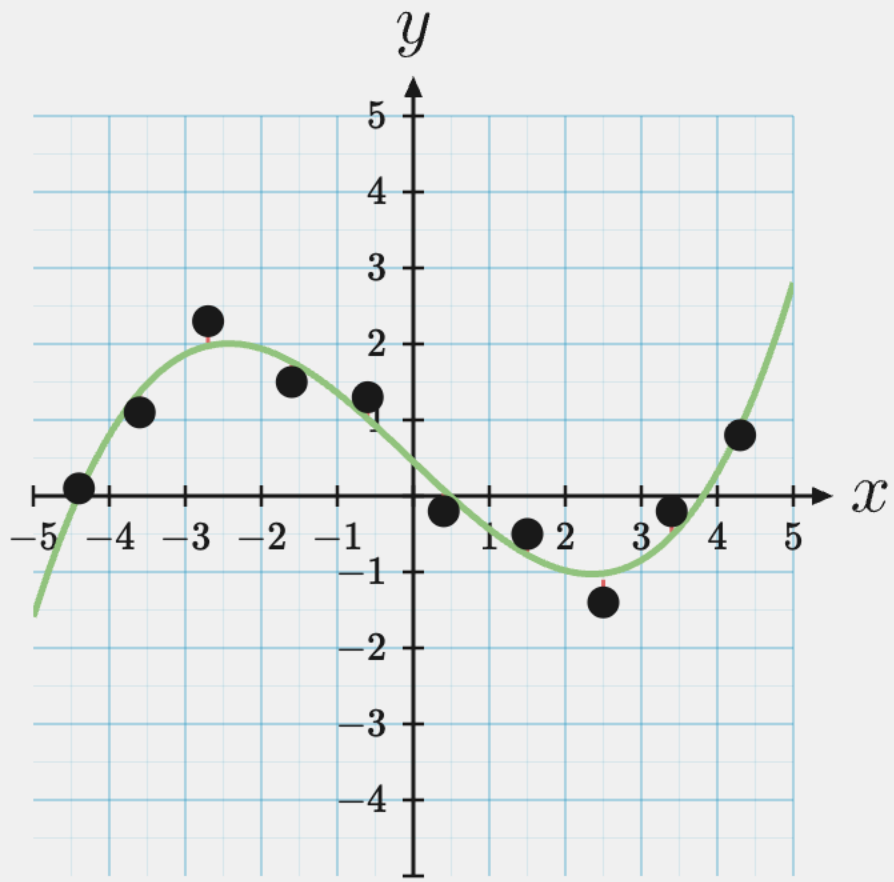
$$\phi_k(x) := [x^k, x^{k-1}, \cdots, x, 1]^\top$$

$$\hat{y} = \phi_k(x)^\top \theta$$

Best parameter:

$$\theta = (\boldsymbol{X}^\top \boldsymbol{X})^{-1} \boldsymbol{X}^\top \boldsymbol{y}$$

Polynomial Basis Expansion



$$\phi_k(x) := [x^k, x^{k-1}, \cdots, x, 1]^\top$$

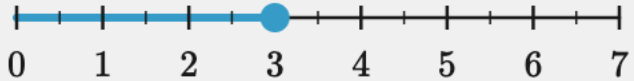
$$\hat{y} = \phi_k(x)^\top \theta$$

$$\theta = \left(\phi_k(\boldsymbol{X})^\top \phi_k(\boldsymbol{X})\right)^{-1} \phi_k(\boldsymbol{X})^\top \boldsymbol{y}$$

$$\boldsymbol{k = 3}, \quad \boldsymbol{\theta \in \mathbb{R}^4}$$

$$\text{training error} = 0.746$$

$$\boldsymbol{k = 3}$$



Regularization

Regularization is a technique used to prevent overfitting by constraining the complexity of a model during training. A prominent way to do this is to penalize the magnitude of the parameters, with a strength λ:

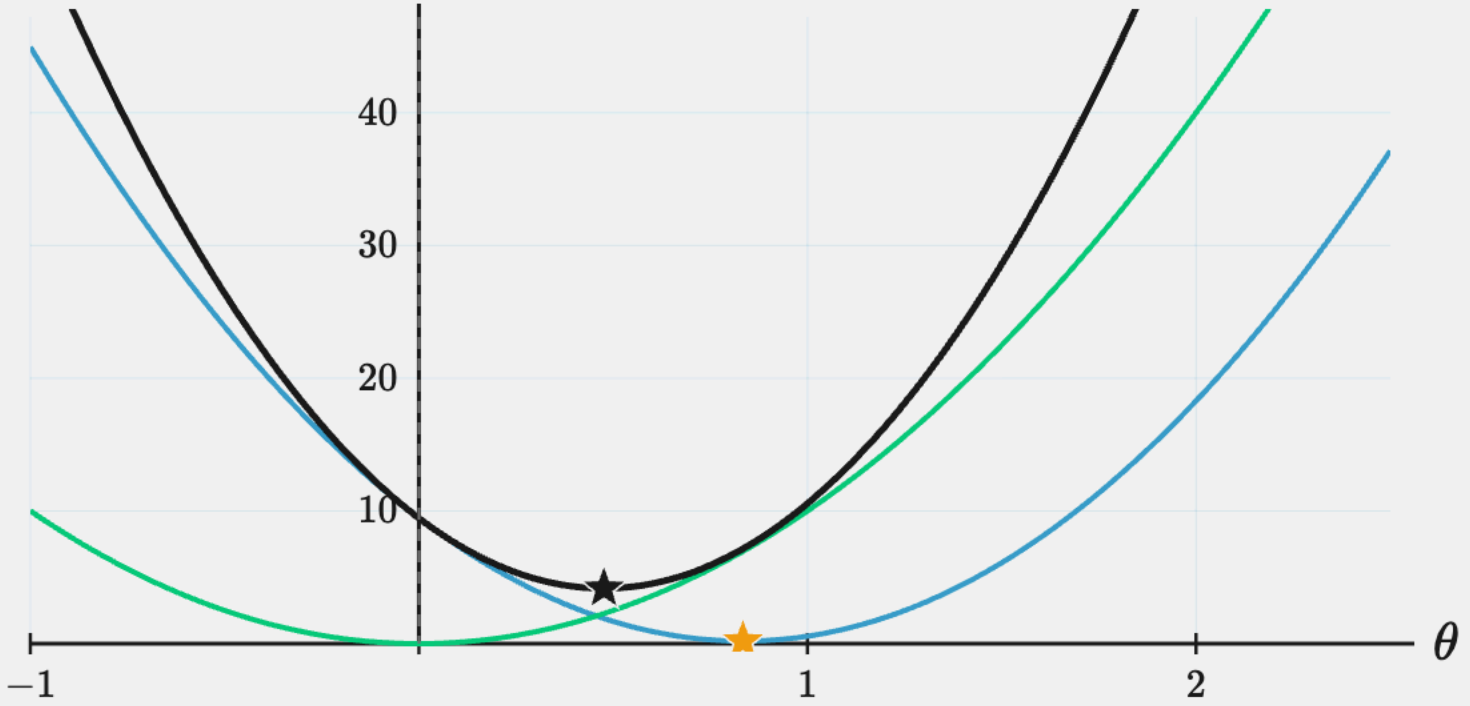
$$\mathcal{L}_{\text{reg}}(\theta, \lambda) := \mathcal{L}(\theta) + \lambda \|\theta\|_2^2$$

↑ λ is what we call a *hyper-parameter*

You will often see this termed *weight decay*. In this lecture we will inspect its effect on the loss landscape of the basis-expanded model, and on the final learned model itself. In our future discussions we will see the very deep theoretical and philosophical underpinnings of this mechanism.

Regularization

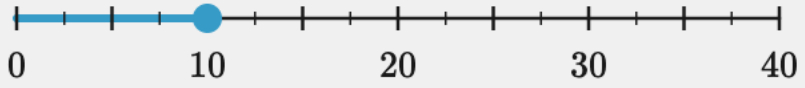
$$\mathcal{L}_{\text{reg}}(\theta, \lambda) = \mathcal{L}(\theta) + \lambda \|\theta\|_2^2$$



data loss $\mathcal{L}(\theta)$
penalty $\lambda\|\theta\|_2^2$
total $\mathcal{L}_{\text{reg}}$
data minimum
total minimum

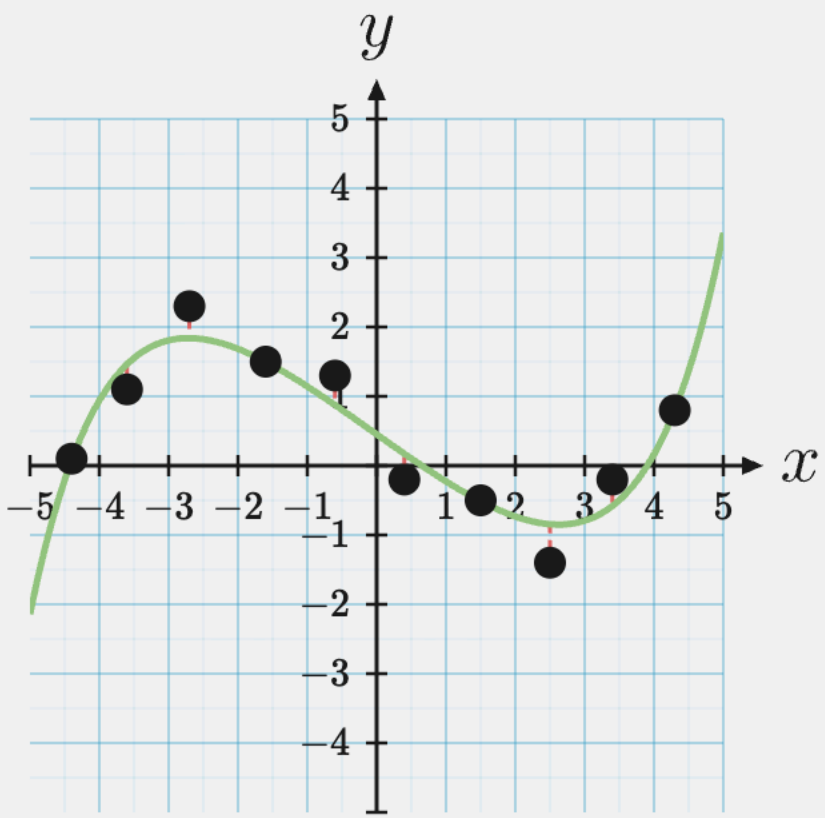
$\theta^\star = 0.83$ (data alone)
$\theta^\star{}_\lambda = 0.48$ (with the penalty)

$\lambda = 10.0$

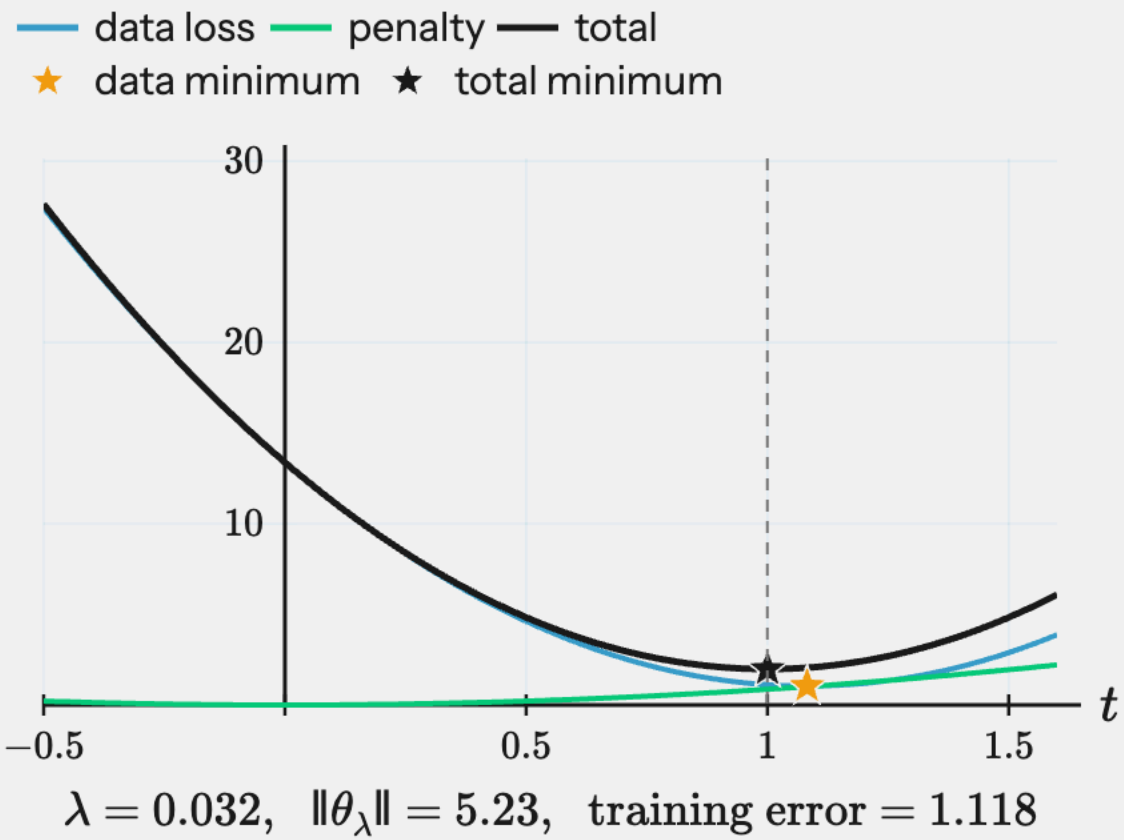


Regularization

$$\hat{y} = \phi_k(x)^\top \theta_\lambda$$

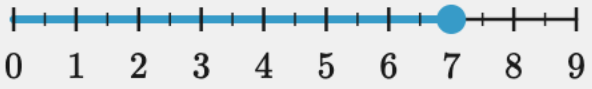


$$\mathcal{L}_{\text{reg}}(t\,\theta_\lambda, \lambda) \quad \text{along} \quad \theta = t\,\theta_\lambda$$

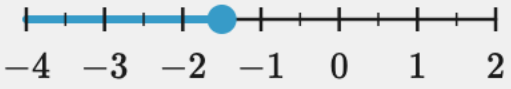


$\lambda = 0.032, \quad \|\theta_\lambda\| = 5.23, \quad \text{training error} = 1.118$

$k = 7$



$log_{10}\,\lambda = -1.50$



Basis expansion is powerful, but has limits

We want to predict the number of times a ship will get damaged in the particular period given some information about the ship

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File Edit Analyze Tables Graphs Curves Vars Help

5	Nom	Nom	Nom	Int	Int
40	TYPE	YEAR	PERIOD	MONTHS	Y
1	b	1965-69	1975-79	20370	53
2	b	1970-74	1960-74	7064	12
3	b	1970-74	1975-79	13099	44
4	b	1975-79	1960-74	0	0
5	b	1975-79	1975-79	7117	18
6	b	1960-64	1960-74	44882	39
7	b	1960-64	1975-79	17176	29
8	b	1965-69	1960-74	28609	58
9	c	1960-64	1960-74	1179	1
10	c	1960-64	1975-79	552	1

Source: https://www.sfu.ca/sasdoc/sashtml/insight/chap17/sect1.htm

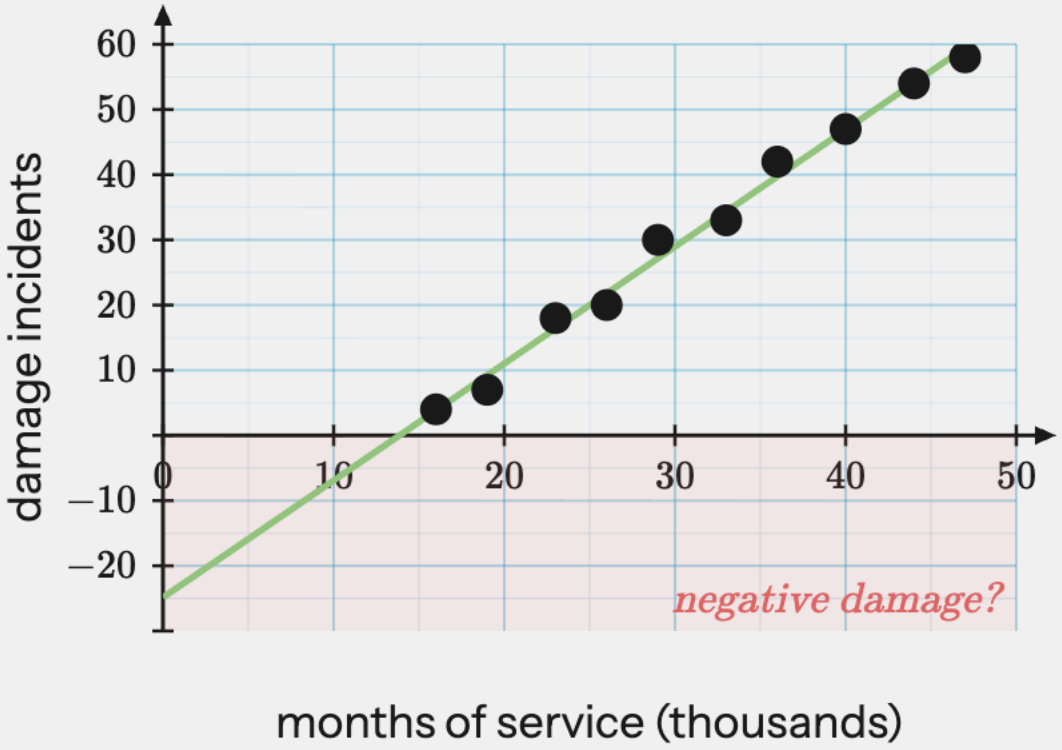
Consider predicting damage to ships:

We want to predict the number of times a ship will get damaged in the particular period given some information about the ship

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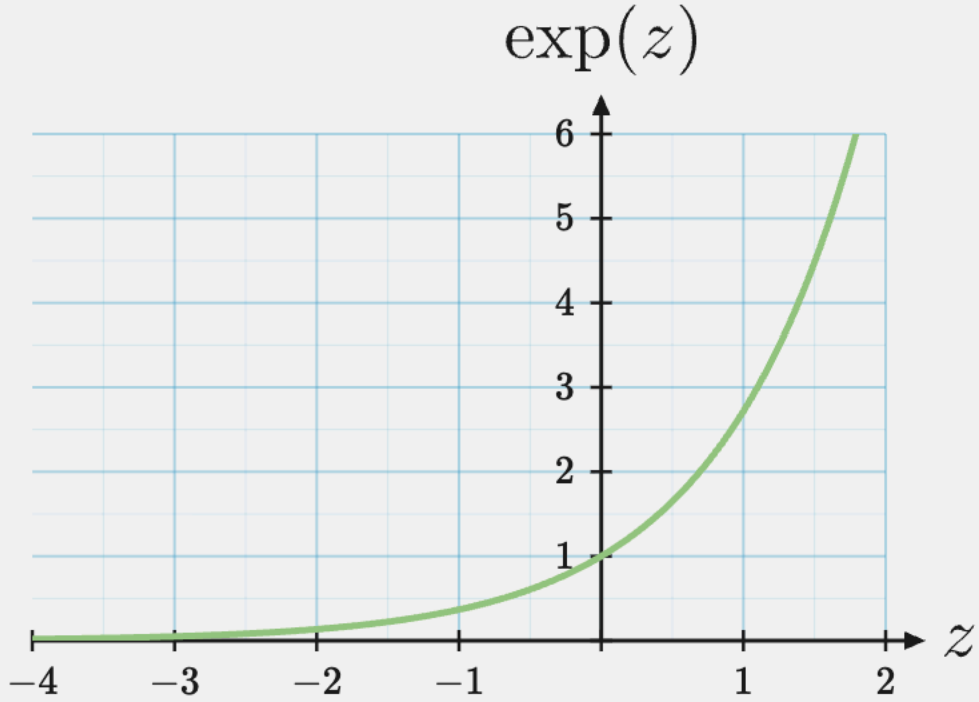


Is linear regression suitable for this situation?

Modifying our model to match our assumptions:

One solution, is to pass our output through a function that is strictly positive:

$$\hat{y} = \exp(\theta^\top \boldsymbol{x}^{(i)})$$



Modifying our model to match our assumptions:

One solution, is to pass our output through a function that is strictly positive:

$$\hat{\boldsymbol{y}} = \exp(\boldsymbol{X}\theta)$$

This yields "poisson regression" which is so-named because of the poisson distribution that we will see in a later lecture.

Applying our previous algorithm:

Today, let's consider fitting this model with the mean squared error:

$$\mathcal{L}(\theta) = \|\hat{\boldsymbol{y}} - \boldsymbol{y}\|_2^2$$

$$= \|\exp(\boldsymbol{X}\theta) - \boldsymbol{y}\|_2^2$$

$$= \sum_{i=1}^{m} \left(\exp(\boldsymbol{X}_{i,:}^\top \theta) - y_i\right)^2$$

$$= (\exp(\boldsymbol{X}\theta) - \boldsymbol{y})^\top (\exp(\boldsymbol{X}\theta) - \boldsymbol{y})$$

Applying our previous algorithm:

Today, let's consider fitting this model with the mean squared error:

$$\mathcal{L}(\theta) = \sum_{i=1}^{m} \left(\exp(\theta^\top \boldsymbol{X}_{i,:}) - y_i \right)^2$$

"Learning" Algorithm:

1. Expand our loss function

Applying our previous algorithm:

Today, let's consider fitting this model with the mean squared error:

$$\mathcal{L}(\theta) = \sum_{i=1}^{m} \left(\exp(\theta^\top \boldsymbol{X}_{i,:}) - y_i \right)^2$$

"Learning" Algorithm:

1. Expand our loss function
2. Take gradient of loss function
3. Set gradient equal to zero, solve for theta

Solving for the gradient

We already have our loss expanded into summation notation, so now let's take the gradient:

$$\mathcal{L}(\theta) = \sum_{i=1}^{m} \left(\exp(\boldsymbol{X}_{i,:}^{\top} \theta) - y_i \right)^2$$

When we have a composition of functions (operations nested) we need the chain rule as in single variable calculus:

$$\left(\exp(\boldsymbol{X}_{i,:}^{\top} \theta) - y_i \right)^2$$

Solving for the gradient: chain rule

When we have a composition of functions (operations nested) we need the chain rule as in single variable calculus:

$$\left(\exp(\boldsymbol{X}_{i,:}^{\top}\theta) - y_i\right)^2$$

Namely, where F is a composition of functions f and g we have:

$$F'(x) = f'\Big(g(x)\Big) \cdot g'(x)$$

Solving for the gradient: chain rule

$$\left(\exp(\boldsymbol{X}_{i,:}^\top \theta) - y_i\right)^2$$

$$f : (\cdot)^2 \qquad\qquad g : \exp(\boldsymbol{X}_{i,:}^\top \theta)$$

$$F'(x) = f'\big(g(x)\big) \cdot g'(x)$$

$$\frac{\partial}{\partial \theta}\left(\exp(\boldsymbol{X}_{i,:}^\top \theta) - y_i\right)^2 = 2\left(\exp(\boldsymbol{X}_{i,:}^\top \theta) - y_i\right) \cdot \frac{\partial}{\partial \theta} \exp(\boldsymbol{X}_{i,:}^\top \theta)$$

Set equal to 0?

$$\frac{\partial}{\partial \theta}\left(\exp(\boldsymbol{X}_{i,:}^\top \theta) - y_i\right)^2$$

$$= 2\left(\exp(\boldsymbol{X}_{i,:}^\top \theta) - y_i\right) \cdot \frac{\partial}{\partial \theta} \exp(\boldsymbol{X}_{i,:}^\top \theta)$$

$$= 2\left(\exp(\boldsymbol{X}_{i,:}^\top \theta) - y_i\right) \cdot \exp(\boldsymbol{X}_{i,:}^\top \theta) \cdot \frac{\partial}{\partial \theta}(\boldsymbol{X}_{i,:}^\top \theta)$$

$$= 2\left(\exp(\boldsymbol{X}_{i,:}^\top \theta) - y_i\right) \cdot \exp(\boldsymbol{X}_{i,:}^\top \theta) \cdot \boldsymbol{X}_{i,:}^\top$$

$$\nabla_\theta \mathcal{L}(\theta) = \sum_{i=1}^{m} 2\left(\exp(\boldsymbol{X}_{i,:}^\top \theta) - y_i\right) \exp(\boldsymbol{X}_{i,:}^\top \theta)\, \boldsymbol{X}_{i,:}^\top$$

$$= 2\left(\left(\exp(\boldsymbol{X}\theta) - \boldsymbol{y}\right) \circ \exp(\boldsymbol{X}\theta)\right)^\top \boldsymbol{X}$$

Why can't we solve for θ?

$$\text{OLS:} \quad 0 = \theta^\top (\boldsymbol{X}^\top \boldsymbol{X}) - \boldsymbol{y}^\top \boldsymbol{X} \quad \Rightarrow \quad \theta = (\boldsymbol{X}^\top \boldsymbol{X})^{-1} \boldsymbol{X}^\top \boldsymbol{y}$$

$$\text{Poisson:} \quad 0 = 2\big((\exp(\boldsymbol{X}\theta) - \boldsymbol{y}) \circ \exp(\boldsymbol{X}\theta)\big)^\top \boldsymbol{X}$$

- For OLS the gradient is **linear** in θ, so $\nabla_\theta \mathcal{L} = 0$ is a linear system: matrix algebra isolates θ.
- Here θ sits inside exp(·), twice, multiplied element-wise: the equation is **non-linear** in θ.
- No rearranging isolates θ, so there is no closed-form solution.

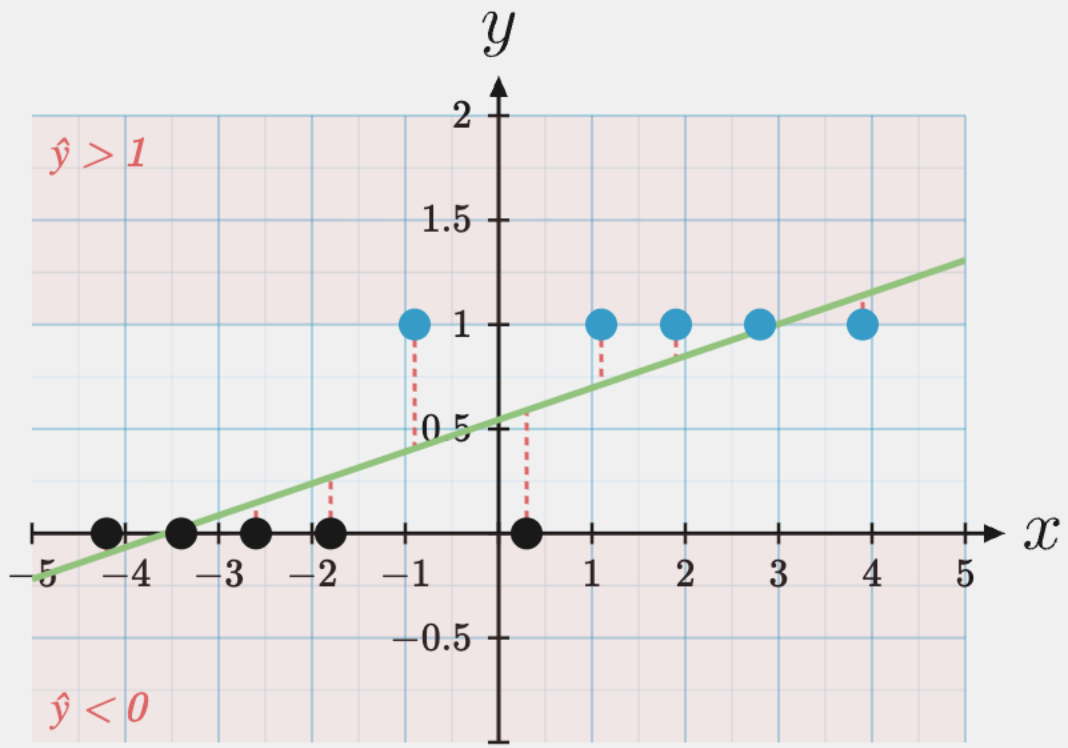
Linear regression for classification?

Another constraint we might consider is given in the classification setting: we don't just want outputs to be positive, we want them to be in (0,1).

*binary classification first



Linear regression for classification?



$$\hat{y} = wx + b$$

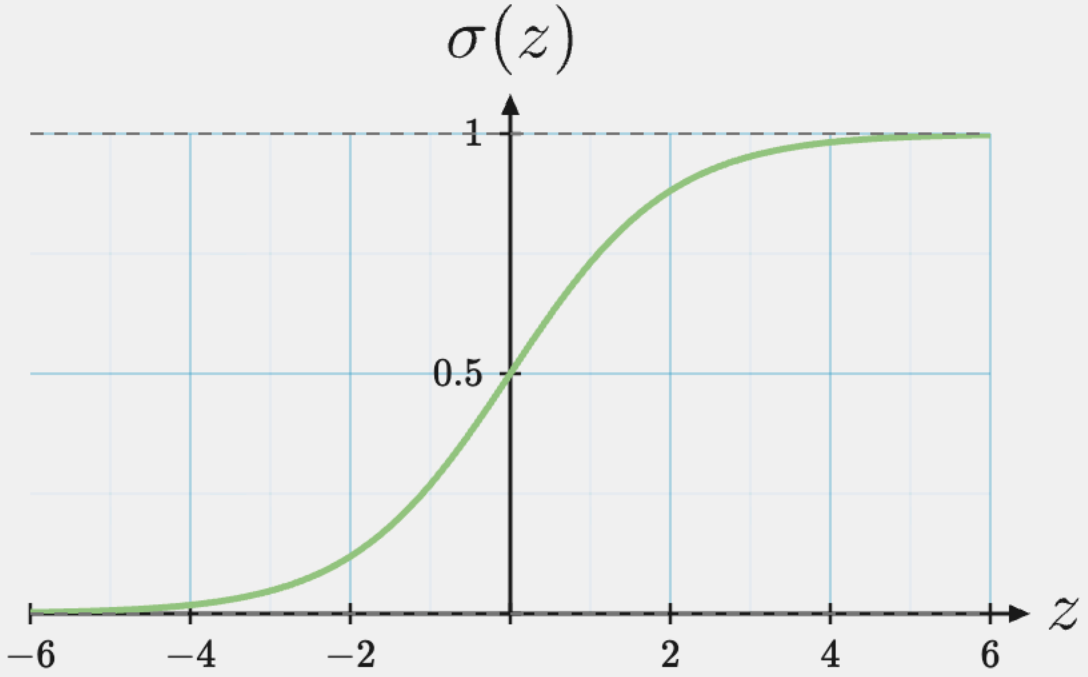
$$w = 0.15, \quad b = 0.54$$

ŷ outside (0, 1) for x < −3.6 and x > 3.0

Linear regression for classification

Another constraint we might consider is given in the classification setting: we don't just want outputs to be positive, we want them to be in (0,1).

$$\hat{y} = \sigma(\theta^\top \boldsymbol{x}^{(i)}) = \frac{1}{1 + \exp(-\theta^\top \boldsymbol{x}^{(i)})}$$



Uniting poisson regression and logistic regression

Generalized Linear Model (GLM):

$$\hat{y} = g^{-1}(\theta^\top \boldsymbol{x})$$

GLM with basis expansion:

$$\hat{y} = g^{-1}\left(\theta^\top \phi(\boldsymbol{x})\right)$$

The function g is called the link function.

Our old algorithm fails

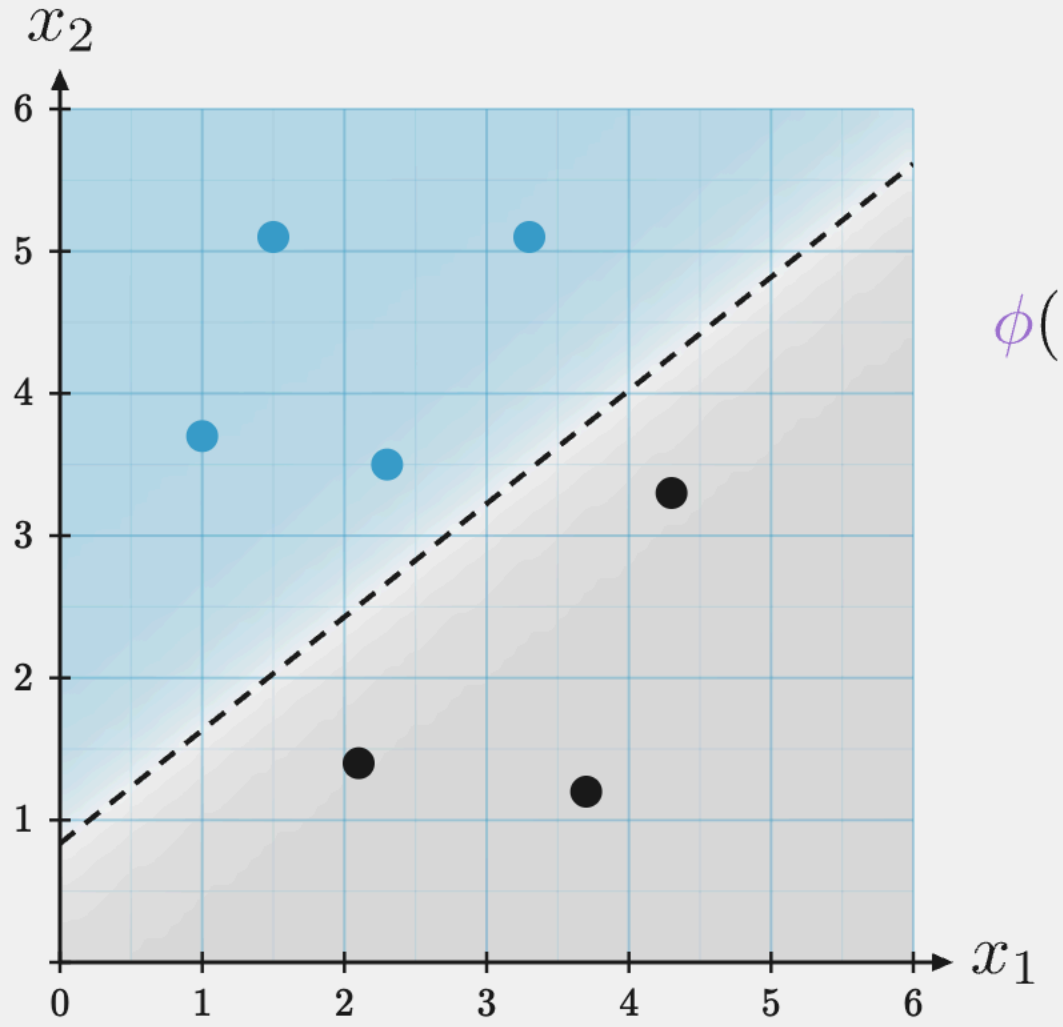
Logistic regression is one where we will also see the derivation of the loss function later:

$$\mathcal{L}(\theta) = -\sum_{i=1}^{m} \left(y^{(i)} \log \left(\sigma(\theta^\top \boldsymbol{x}^{(i)}) \right) + (1 - y^{(i)}) \log \left(1 - \sigma(\theta^\top \boldsymbol{x}^{(i)}) \right) \right)$$

But we can still do the first two steps!

1. Expand our loss function
2. Take gradient of loss function
3. Set gradient equal to zero, solve for theta

A preview: logistic regression, live



$$\hat{y} = \sigma\left(\theta^\top \phi(\boldsymbol{x})\right)$$

$$\phi(\boldsymbol{x}) = [x_1, x_2, 1]^\top, \qquad \sigma^{(i)} := \sigma\left(\theta^\top \phi(\boldsymbol{x}^{(i)})\right)$$

$$\mathcal{L}(\theta) = -\sum_{i=1}^{m} \left(y^{(i)} \log \sigma^{(i)} + (1 - y^{(i)}) \log(1 - \sigma^{(i)}) \right)$$

$$\textbf{iteration } 124, \;\; \mathcal{L}(\theta) = 0.045$$

$$\theta = [-1.92,\; 2.41,\; -2.02]^\top$$

θ found by gradient descent — next lecture

Practice Exam Question

1 **Deriving the Ridge Regression Estimator**

Let $\mathbf{X} \in \mathbb{R}^{m \times n}$ be a design matrix, let $\mathbf{y} \in \mathbb{R}^m$ be the vector of responses, and let $\boldsymbol{\theta} \in \mathbb{R}^n$ be the parameter vector. For a fixed hyper-parameter $\lambda > 0$, the ridge regression objective is

$$\mathcal{L}_{\text{ridge}}(\boldsymbol{\theta}) \;=\; \frac{1}{m}\,\|\mathbf{y} - \mathbf{X}\boldsymbol{\theta}\|_2^2 \;+\; \lambda\,\|\boldsymbol{\theta}\|_2^2.$$

a Write $\mathcal{L}_{\text{ridge}}(\boldsymbol{\theta})$ in index (summation) notation, using only scalar entries X_{ij}, y_i and θ_j.

b Differentiate $\mathcal{L}_{\text{ridge}}$ with respect to an arbitrary parameter θ_k, collect your result into the full gradient $\nabla_{\boldsymbol{\theta}}\mathcal{L}_{\text{ridge}}$, and solve for the optimal parameter vector $\boldsymbol{\theta}^\star$. Justify that the matrix you invert is invertible and that your stationary point is a minimum.

c In lectures we saw that the effect of regularization was to bring the magnitude of the learned parameters down. Identify the term in your answer to part (ii) that is responsible for this, and explain why it results in a parameter vector of lower magnitude than the ordinary least squares estimate $\boldsymbol{\theta}_{\text{OLS}} = (\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top\mathbf{y}$.

The three parts carry, respectively, 20%, 50%, and 30% of the marks.

Feedback



https://forms.gle/ymy7egC61k7FkhmdA